

OSSP PRACTICAL 3

- created a C program using the fork() system call.
- The child process displayed the PID and PPID of both the parent and child processes to identify them separately.

```
vida@LAPTOP-4B7UUAUH:~/os_lab$ nano prac3.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc prac3.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
parent process
parent pid is : 665
parent's parent pid is : 304
child process success
Child PID is : 666
Parent pid is : 665
```

- During the execution of the program, the ps command was used to view the processes currently running in the terminal.

```
vida@LAPTOP-4B7UUAUH:~/os_lab$ nano prac3.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc prac3.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
parent process
parent pid is : 665
parent's parent pid is : 304
child process success
Child PID is : 666
Parent pid is : 665
vida@LAPTOP-4B7UUAUH:~/os_lab$ ps
  PID TTY          TIME CMD
  304 pts/0        00:00:00 bash
  666 pts/0        00:00:58 a.out
  682 pts/0        00:00:00 ps
```

- Used the top command to continuously monitor the processes while they were executing.

```
vida@LAPTOP-4B7UUAUH:~/os_lab$ top
top - 18:54:56 up 19 min, 1 user, load average: 0.95, 0.45, 0.18
Tasks: 25 total, 2 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 8.3 us, 0.0 sy, 0.0 ni, 91.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7788.6 total, 7187.3 free, 535.1 used, 217.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7253.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+    COMMAND
  666 vida       20   0    2768    1064    968 R   100.0   0.0   2:48.12 a.out
    1 root        20   0   24556   15488   11616 S    0.0   0.2   0:00.56 systemd
    2 root        20   0    3180    2208    2072 S    0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2048    1940 S    0.0   0.0   0:00.00 init
   47 root       19  -1   50400   16764   15484 S    0.0   0.2   0:00.14 systemd-journal
   79 systemd+    20   0   22540   14628   12008 S    0.0   0.2   0:00.05 systemd-resolve
   87 root        20   0   35364   12316    9216 S    0.0   0.2   0:00.50 systemd-udev

top - 18:56:01 up 20 min, 1 user, load average: 0.99, 0.56, 0.24
Tasks: 25 total, 2 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 8.2 us, 0.1 sy, 0.0 ni, 91.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7788.6 total, 7187.1 free, 535.3 used, 217.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7253.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+    COMMAND
  666 vida       20   0    2768    1064    968 R   100.0   0.0   3:53.20 a.out
    1 root        20   0   24556   15488   11616 S    0.0   0.2   0:00.56 systemd
    2 root        20   0    3180    2208    2072 S    0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2048    1940 S    0.0   0.0   0:00.00 init
   47 root       19  -1   50400   16764   15484 S    0.0   0.2   0:00.14 systemd-journal
   79 systemd+    20   0   22540   14628   12008 S    0.0   0.2   0:00.05 systemd-resolve
   87 root        20   0   35364   12316    9216 S    0.0   0.2   0:00.51 systemd-udev

top - 18:57:19 up 22 min, 1 user, load average: 1.00, 0.67, 0.30
Tasks: 25 total, 2 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 8.3 us, 0.0 sy, 0.0 ni, 91.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7788.6 total, 7185.8 free, 536.6 used, 217.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7252.1 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+    COMMAND
  666 vida       20   0    2768    1064    968 R   100.0   0.0   5:11.39 a.out
    1 root        20   0   24556   15488   11616 S    0.0   0.2   0:00.56 systemd
    2 root        20   0    3180    2208    2072 S    0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2048    1940 S    0.0   0.0   0:00.00 init
   47 root       19  -1   50400   16764   15484 S    0.0   0.2   0:00.14 systemd-journal
   79 systemd+    20   0   22540   14628   12008 S    0.0   0.2   0:00.05 systemd-resolve
   87 root        20   0   35364   12316    9216 S    0.0   0.2   0:00.52 systemd-udev
  131 root        20   0    2888    1944    1820 S    0.0   0.0   0:00.02 chronyd-starter
  135 root        20   0   4472    3152    2904 S    0.0   0.0   0:00.00 cron
  142 message+    20   0    8812    5384    4720 S    0.0   0.1   0:00.04 dbus-daemon
  153 root        20   0   44092   29804   14664 S    0.0   0.4   0:00.18 networkd-dispat
  157 root        20   0   18436    9428    8184 S    0.0   0.1   0:00.06 systemd-logind
  171 syslog      20   0   220548   5848    4540 S    0.0   0.1   0:00.04 rsyslogd
  226 _chrony      20   0   20420    6336   5560 S    0.0   0.1   0:00.07 chronyd
  241 _chrony      20   0   12092    2468    1824 S    0.0   0.0   0:00.01 chronyd
  266 root        20   0   123456   32644   18032 S    0.0   0.4   0:00.11 unattended-upgr
  269 root        20   0    5492    2704    2524 S    0.0   0.0   0:00.01 agetty
  298 root        20   0    3184    1112     980 S    0.0   0.0   0:00.00 SessionLeader
```

- Used the /proc filesystem to inspect the process with PID 666, which was the a.out process identified using the ps command.

```
vida@LAPTOP-4B7UUAUH:~/os_lab$ ls /proc/666
arch_status  cpuset      ksm_merging_pages  mountinfo  oom_score_adj  sessionid  syscall
attr         cwd         ksm_stat           mounts     pagemap        setgroups  task
auxv         environ    latency           mountstats patch_state    smaps      timers_offsets
cgroup       exe        limits            net        personality    smaps_rollup  timers
clear_refs   fd         loginuid          ns         projid_map     stack      timerslack_ns
cmdline      fdinfo     map_files         numa_maps  root           stat       uid_map
comm         gid_map    maps              oom_adj    sched          statm      wchan
coredump_filter io         mem              oom_score  schedstat     status
```